

## OSSP (Operating Systems And Systems Programming) - 25CS2104E

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### WEEK 2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

CODE :

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
int main() {
```

```
    char source[100], destination[100];
```

```
    char buffer[1024];
```

```
    int src_fd, dest_fd;
```

```
    ssize_t n;
```

```
    printf("Enter source file name: ");
```

```
    scanf("%99s", source);
```

```
    printf("Enter destination file name: ");
```

```
    scanf("%99s", destination);
```

```
    src_fd = open(source, O_RDONLY);
```

```
    dest_fd = open(destination, O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```
    while ((n = read(src_fd, buffer, sizeof(buffer))) > 0) {
```

```
    write(dest_fd, buffer, n);  
}  
close(src_fd);  
close(dest_fd);  
printf("File copied successfully.\n");  
return 0;  
}
```

OUTPUT :

```
pothubhavana@DESKTOP-16SJJUA:~$ nano source.txt  
pothubhavana@DESKTOP-16SJJUA:~$ nano week2.c  
pothubhavana@DESKTOP-16SJJUA:~$ gcc week2.c  
pothubhavana@DESKTOP-16SJJUA:~$ ./a.out  
Enter source file name: operating systems  
Enter destination file name: File copied successfully.
```